CIPHER: Conformer-based Inference of Phonemes from High-density EEG Representations

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Abstract. Decoding speech information from scalp EEG remains difficult due to low SNR and spatial blurring. We present **CIPHER** (Conformer-based Inference of Phonemes from High-density EEG Representations), a dual-pathway model using (i) ERP features and (ii) broadband DDA coefficients. On OpenNeuro ds006104 (24 participants, two studies with concurrent TMS), binary articulatory tasks reach near-ceiling performance but are highly confound-vulnerable (acoustic onset separability and TMS-target blocking). On the primary 11-class CVC phoneme task under full Study 2 LOSO (16 held-out subjects), performance is substantially lower (real-word WER: ERP 0.671 ± 0.080, DDA 0.688 ± 0.096), indicating limited fine-grained discriminability. We therefore position this work as a benchmark and feature-comparison study rather than an EEG-to-text system, and we constrain neural-representation claims to confound-controlled evidence.

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Dedication

This work is dedicated to the memory of my grandfather, **Late Tarala Narsingha Rao**, who lived with a lifelong neural disorder that severely limited his ability to communicate. From early childhood, he faced profound difficulty expressing his thoughts, needs, and pain. Over time, his ability to communicate diminished further, until he was no longer able to convey even discomfort while enduring significant internal suffering.

Witnessing this silence shaped my understanding of the human cost of lost communication. This work is guided by the belief that restoring even a fragile channel of expression can transform silence into dignity, isolation into connection, and agency for individuals who would otherwise remain unheard.

Introduction

The ability to decode speech from neural activity has profound implications for assistive communication technologies, particularly for individuals with severe motor impairments such as locked-in syndrome or amyotrophic lateral sclerosis (ALS). While intracortical recordings have achieved remarkable decoding performance—including real-time sentence-level speech synthesis [1–3]—the clinical invasiveness of implanted electrode arrays severely limits their scalability. Scalp EEG offers a non-invasive alternative with millisecond temporal resolution, but suffers from volume conduction, low signal-to-noise ratio, and spatial blurring, making fine-grained speech decoding substantially more challenging.

Prior work on EEG-based speech decoding has largely relied on either traditional event-related potential (ERP) analysis or spectral feature extraction [4, 5]. ERPs capture the phase-locked cortical response to stimulus events (e.g., the auditory N1/P2 complex, the speech-selective N200), but by averaging across trials, they discard the trial-by-trial temporal dynamics that carry discriminative information. Conversely, approaches based on spectral or time-frequency features can capture sustained oscillatory activity but may miss the transient, nonlinear dynamics characteristic of speech processing.

In this work, we propose a dual-pathway approach that computes two complementary feature representations from the same raw EEG recording:

1. **Path A (ERP):** Standard event-related preprocessing — downsampling to 256 Hz, band-pass filtering (0.5–40 Hz), common average re-referencing, ICA-based artifact rejection — followed by stimulus-locked epoching with amplitude-based rejection.
2. **Path B (DDA):** Delay differential analysis [6] on the raw 2048 Hz signal, computing the coefficients of a three-term nonlinear dynamical model via Cramér's rule in sliding windows, then epoching the resulting coefficient time series.

Both feature streams are independently fed into identical Conformer-based architectures [7], and an ensemble of the two models is also evaluated. Our contributions are:

- **A Conformer-based EEG adaptation study:** we adapt and evaluate a Conformer-style encoder for EEG decoding, and ablations show that SE channel attention is the most consistent component-level contributor across tasks.
- **Multi-task joint training** across phoneme identity, place of articulation, manner, and voicing—enabling shared representation learning across articulatory feature hierarchies.
- **Dual-pathway feature extraction** (ERP + DDA) with late logit-averaging fusion, demonstrating that nonlinear dynamical features provide complementary discriminative information.
- **Broad evaluation** across six classification tasks, three TMS conditions, two studies with cross-dataset transfer, and lexicality analyses.
- **A transparent interpretation of limits:** we explicitly identify acoustic and TMS-design confounds that can dominate binary-task performance, and we constrain claims accordingly.

Scope statement. This study evaluates stimulus-locked EEG classification under known confound risks. It does not demonstrate open-vocabulary EEG-to-text decoding.

Code availability. All code, preprocessing pipelines, and training configurations are publicly available at `https://github.com/Varshith-O/CIPHER`.

Related Work

EEG-Based Speech Decoding

Early BCI work focused on imagined or attempted speech classification using discriminative features from select EEG channels [8]. More recent deep learning approaches include CNNs [9], LSTMs, and Transformer variants [10] applied to EEG spectrograms or raw signals. Notably, EEGNet [11] introduced depthwise separable convolutions tailored to EEG's spatial-temporal structure, while DeWave [12] explored translating EEG to text via discrete codex representation. Recent non-invasive speech-decoding studies also emphasize strict benchmark framing and careful interpretation of decoding claims [4]. Accordingly, we treat EEGNet [11] and ShallowConvNet [9] as direct architectural comparators in our matched-split baseline table.

Delay Differential Analysis

DDA, introduced by Lainscsek and Sejnowski [6], models a time series as a nonlinear delay-differential equation:

$$\dot{x}(t) = a_1 x(t-\tau_1) + a_2 x(t-\tau_2) + a_3 x(t-\tau_1)^2.$$

The coefficients (a_1, a_2, a_3) are solved via least squares (Cramér's rule) in sliding windows and characterize the system's attractor geometry. DDA has shown promise for EEG classification in epilepsy detection [13] and cognitive state monitoring, but has not been systematically compared against ERPs for speech decoding.

Conformer Networks

The Conformer architecture [7] combines self-attention for modeling global dependencies with convolution for capturing local patterns, using a macaron-style sandwich: FFN $\rightarrow$ MHSA $\rightarrow$ ConvModule $\rightarrow$ FFN. Originally developed for automatic speech recognition (ASR) on audio, Conformers have shown state-of-the-art results on sequential pattern recognition tasks. We adapt this architecture for EEG by replacing the acoustic front-end with a multi-scale convolutional feature extractor tailored to the spectral characteristics of neural signals.

TMS and Speech Motor Theory

The motor theory of speech perception posits that perceiving speech involves simulating articulatory gestures in the motor cortex [14]. Transcranial magnetic stimulation (TMS) to speech motor areas during a phoneme discrimination task provides a causal intervention: stimulating lip motor cortex should selectively facilitate discrimination of bilabial phonemes (/b/, /p/), while tongue motor cortex stimulation should facilitate alveolar phonemes (/d/, /t/, /s/, /z/) [15]. This creates a unique opportunity to test whether neural decoding performance is modulated by motor cortex excitability.

Methods

Dataset

We use the publicly available OpenNeuro dataset ds006104 [16], comprising two related studies:

Table 1: Dataset overview.

	Study 1 (2019)	**Study 2 (2021)**
Participants	8 (sub-P01–P08)	16 (sub-S01–S16)
Session	ses-01	ses-02
Tasks	CV and VC phoneme pairs	Single phonemes, CV/VC, CVC
Phonemes	6 consonants + 5 vowels	Same
EEG System	64-ch BioSemi ActiveTwo	Same
Sampling Rate	2048 Hz (raw)	2048 Hz (raw)
TMS	Paired-pulse (110% rMT) to LipM1, TongueM1, control	Same

Phoneme Classification Tasks. We define six classification targets:

1. **Phoneme Identity** (11 classes): Individual phoneme recognition — *a, b, d, e, i, o, p, s, t, u, z.*
2. **Place of Articulation** (2 classes): *alveolar* (d, t, s, z) vs. *bilabial* (b, p).
3. **Manner of Articulation** (2 classes): *fricative* (s, z) vs. *stop* (b, d, p, t).
4. **Voicing** (2 classes): *voiced* (b, d, z) vs. *unvoiced* (p, s, t).
5. **Category** (2 classes): *consonant* vs. *vowel.*
6. **Complexity** (3 classes): *single phoneme* vs. *diphone* (CV/VC) vs. *triphone* (CVC).

Data Splits. The primary evaluation split is full Study 2 leave-one-subject-out (LOSO; 16 held-out subjects), used for the main phoneme-level WER results and control analyses. We also report a fixed Study 2 split (13 train, 3 validation: S04, S09, S14) for comparability with earlier development runs and for matched-split baseline tables, with Study 1 (all 8 subjects) as held-out cross-study test for transfer checks. To reduce small-n validation fragility for architectural conclusions, we additionally run an expanded Study 2 split (S01–S08 train, S09–S16 validation; 8 held-out subjects) for ablation robustness.

Preprocessing

Path A: ERP Feature Extraction

1. **Loading:** Prefer EEGLAB-cleaned derivatives (with TMS artifact removal) when available; otherwise load raw EDF files.
2. **Channel Selection:** Drop non-EEG channels (Status, EOG, BIP, EMG, mastoid references). Set all remaining channels to EEG type with standard 10-20 montage.
3. **Resampling:** Downsample to 256 Hz.
4. **Filtering:** Notch filter at 50/60 Hz (power-line artifact rejection, 2 Hz bandwidth), followed by FIR bandpass at 0.5–40 Hz (Hamming window).
5. **Re-referencing:** Common average reference (CAR).
6. **ICA Artifact Rejection:** FastICA with up to 15 components. Automatic identification of EOG-correlated components (via frontal channel correlation) and muscle artifact components (via high-frequency power). At most one-third of components are rejected.
7. **Epoching:** Stimulus-locked epochs from $t = -200$ ms to $t = +800$ ms. Baseline correction using the pre-stimulus window ($[-200, 0]$ ms). Amplitude rejection threshold: 150 μV

(disabled when using EEGLAB-cleaned data, where TMS artifacts are already removed).

Output: Epochs of shape $(T_{\text{erp}} \times C)$, where $T_{\text{erp}} = 256 \times 1.0 = 256$ time points and C is the number of retained EEG channels (typically 60–64).

Path B: DDA Feature Extraction

1. **Loading:** Raw 2048 Hz data (or EEGLAB-cleaned). No filtering applied—DDA operates on the broadband signal.
2. **DDA Computation:** For each EEG channel, compute the delay-differential coefficients (a_1, a_2, a_3) in sliding windows of length 60 samples (≈29.3 ms at 2048 Hz) with a shift of 2 samples (≈0.98 ms), using two time delays $\tau_1 = 6$ samples (≈2.93 ms) and $\tau_2 = 16$ samples (≈7.81 ms).

   The DDA model for each window is:

   $$\dot{x}(t) \approx a_1\, x(t-\tau_1) + a_2\, x(t-\tau_2) + a_3\, x(t-\tau_1)^2$$

   The derivative $\dot{x}(t)$ is estimated via central differences: $\dot{x}(k) = \frac{x(k+1)-x(k-1)}{2\Delta t}$. The coefficients are computed via Cramér's rule on the 3×3 normal equation system, with per-window z-score normalization. When the Numba JIT compiler is available, all channels are processed in parallel via `numba.prange`.
3. **Epoching:** DDA window centres are aligned to stimulus onsets; same temporal window as ERPs ($[-200, +800]$ ms).
4. **Temporal Stride:** A stride of 4 is applied during dataset loading to reduce the DDA sequence length.

Output: Epochs of shape $(T_{\text{dda}} \times C \times 3)$, reshaped to $(T_{\text{dda}} \times 3C)$ for model input, where $T_{\text{dda}} \approx 250$ windows (after stride of 4) and C is the channel count.

Model Architecture

Figure 1 shows the full dual-pathway pipeline. Both ERP and DDA feature streams pass through the same Conformer encoder before branching into task-specific classification heads.

Multi-Scale Convolutional Front-End

Inspired by EEGNet and multi-scale temporal convolutions, the front-end consists of three parallel 1D convolutional branches with kernel sizes $k \in \{3, 7, 15\}$, designed to capture neural dynamics at different temporal granularities (fine transients, evoked response components, and slow oscillatory patterns respectively). Each branch applies:

$$\text{Conv1D}(k) \rightarrow \text{BatchNorm} \rightarrow \text{GELU} \rightarrow \text{Dropout}$$

The three branch outputs are concatenated along the channel dimension (yielding $3 \times C_{\text{conv}} = 192$ channels for $C_{\text{conv}} = 64$), then projected through a linear layer and processed by a **Squeeze-and-Excitation (SE) block** [17] for adaptive channel attention:

$$\text{SE}(x) = x \odot \sigma\big(W_2\, \text{GELU}(W_1\, \text{GAP}(x))\big)$$

where GAP denotes global average pooling over the temporal dimension and σ is the sigmoid function. The SE reduction ratio is 4.

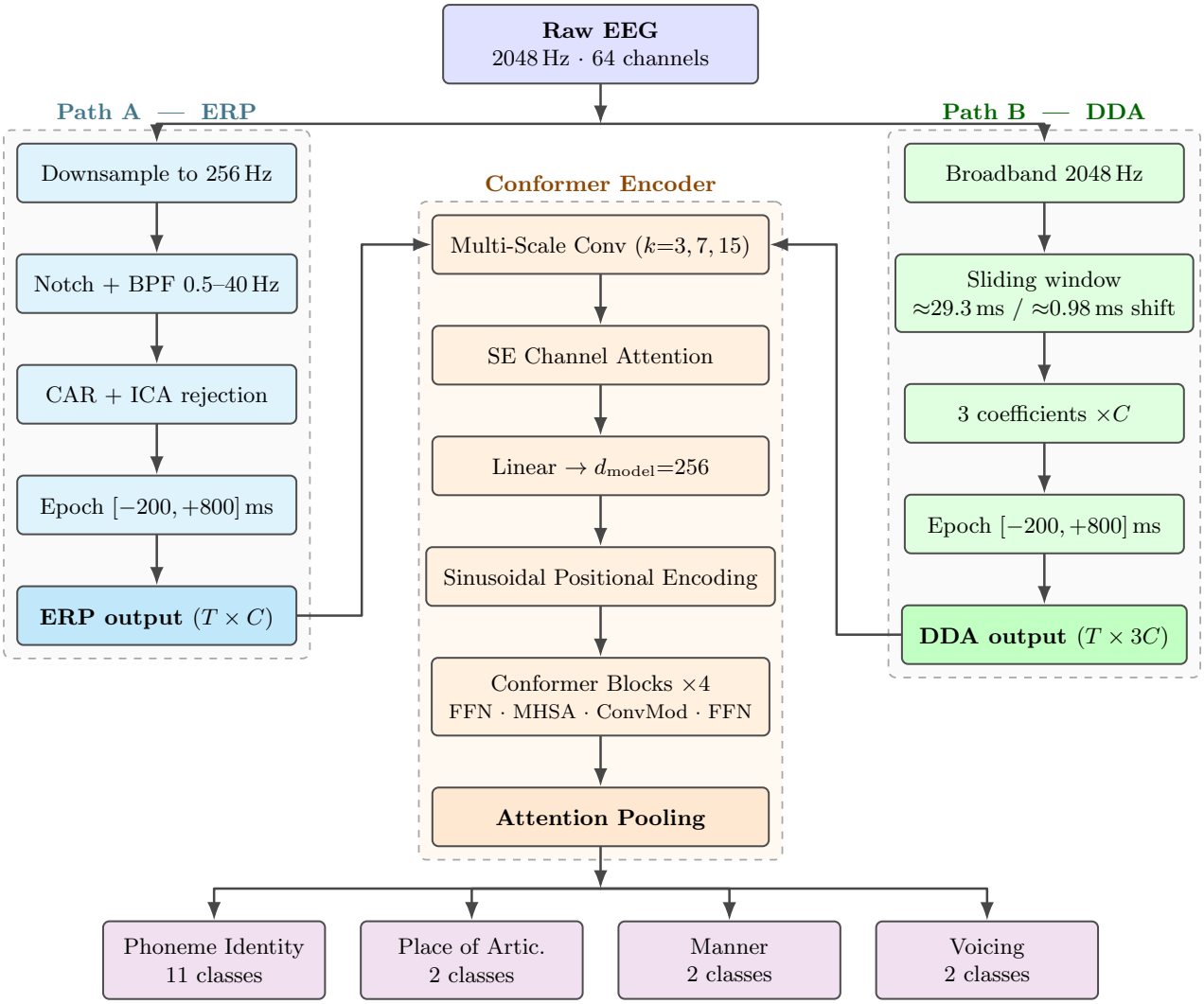

Figure 1: CIPHER dual-pathway architecture. Raw EEG is processed in parallel through an ERP pathway (narrowband, phase-locked dynamics) and a DDA pathway (broadband nonlinear dynamical coefficients), fused by a shared Conformer encoder, and decoded by four task-specific classification heads.

Conformer Encoder

The projected features ($d_{\text{model}} = 256$) are augmented with sinusoidal positional encoding and passed through $N = 4$ stacked Conformer blocks. Each block follows the macaron structure:

$$
\begin{aligned}
x &\leftarrow x + \tfrac{1}{2}\,\text{FFN}_1(\text{LN}(x)) \\
x &\leftarrow x + \text{MHSA}(\text{LN}(x)) \\
x &\leftarrow x + \text{ConvModule}(x) \\
x &\leftarrow x + \tfrac{1}{2}\,\text{FFN}_2(\text{LN}(x)) \\
x &\leftarrow \text{LN}(x)
\end{aligned}
$$

Multi-Head Self-Attention (MHSA): 8 attention heads with dropout.

ConvModule: LayerNorm → Pointwise Conv (×2 expansion) → GLU gating → Depthwise Separable Conv1D (kernel 15, groups = d_{model}) → BatchNorm → SiLU → Pointwise Conv → Dropout + Residual.

Stochastic Depth (DropPath): Drop path rate increases linearly from 0 to 0.05 across blocks, regularizing the deeper layers more aggressively.

Feed-Forward Networks (FFN): Expansion factor 4, GELU activation, with dropout.

Attention Pooling and Classification Heads

A learned query vector $q \in \mathbb{R}^{d_{\text{model}}}$ computes attention weights over the temporal sequence:

$$\alpha_t = \frac{\exp(h_t^\top q)}{\sum_{t'} \exp(h_{t'}^\top q)}, \qquad z = \sum_t \alpha_t h_t$$

The pooled representation z is fed into task-specific three-layer classification heads:

$$\text{Head}_k(z) = W_3 \,\text{GELU}(W_2 \,\text{GELU}(W_1 \,\text{LN}(z)))$$

with intermediate dimensions $256 \to 128$, progressive dropout (heavier in the first layer), and LayerNorm at the input. In multi-task mode, all four articulatory heads (phoneme identity, place, manner, voicing) share the encoder and are trained jointly.

Optional CTC Head

For the phoneme identity task on triphone (CVC) data, an optional Connectionist Temporal Classification (CTC) [18] head produces frame-level phoneme posteriors:

$$\text{CTC}(h_t) = \text{Linear}(\text{LN}(h_t)) \in \mathbb{R}^{|\mathcal{V}|+1}$$

where $|\mathcal{V}| = 11$ phonemes plus a blank token. The CTC loss is weighted by $\lambda_{\text{ctc}} = 0.1$ and added to the classification loss.

Ensemble (ERP + DDA)

An ensemble model averages the logits from independently trained ERP and DDA models:

$$\hat{y}_{\text{ens}} = \arg\max \tfrac{1}{2}\big(\text{logits}_{\text{erp}} + \text{logits}_{\text{dda}}\big)$$

Training Procedure

Loss Function. Label-smoothing cross-entropy with $\epsilon = 0.1$:

$$\mathcal{L}_{\text{LS}} = -\sum_c \tilde{y}_c \log p_c, \qquad \tilde{y}_c = (1-\epsilon)\mathbb{1}[c = y] + \frac{\epsilon}{C}$$

with inverse-frequency class weighting (clipped to $[0.25, 4.0]$) to handle class imbalance.

Mixup Augmentation. During the first 90% of training epochs, mixup [19] is applied with $\alpha = 0.1$:

$$\tilde{x} = \lambda x_i + (1-\lambda) x_j, \qquad \lambda \sim \text{Beta}(\alpha, \alpha)$$

with a dedicated `MixupLabelSmoothingCE` loss for soft targets.

Data Augmentation. Training-time augmentation includes: (1) Gaussian noise injection ($\sigma = 0.02 \times$ signal std); (2) Channel dropout (5–10% of channels zeroed); (3) Random temporal shift (± 5 ERP samples / ± 20 DDA samples); (4) SpecAugment-style time masking (5–10% of time steps); (5) Amplitude scaling (uniform $[0.85, 1.15]$).

Normalization. Per-sample z-score normalization over the temporal dimension for each feature channel.

Optimizer. AdamW with $\beta = (0.9, 0.98)$, learning rate 5×10^{-4}, weight decay 10^{-4}.

Learning Rate Schedule. Cosine annealing with 10-epoch linear warmup:

$$\eta(t) = \begin{cases} \eta_{\max} \cdot \dfrac{t}{T_w} & t < T_w \\ \eta_{\min} + \dfrac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos\left(\pi \cdot \dfrac{t - T_w}{T - T_w}\right)\right) & t \geq T_w \end{cases}$$

Class Balancing. A `WeightedRandomSampler` ensures balanced class representation in each training batch.

Early Stopping. Training runs for up to 150 epochs with patience of 30 epochs on validation accuracy. Gradient clipping at max norm 1.0. Mixed-precision training (AMP) is used when a GPU is available.

Multi-Task Training. When enabled, losses from all four heads are averaged equally:

$$\mathcal{L}_{\text{MT}} = \frac{1}{4} \sum_{k \in \{\text{phon,place,manner,voicing}\}} \mathcal{L}_k$$

All reported results use a single frozen configuration selected on 11-class Study 2 LOSO phoneme WER prior to evaluation.

Evaluation Protocol

We conduct eight complementary analyses, interpreted using the evidence-tier scheme in Section 4:

1. **Cross-Dataset Evaluation (secondary):** Accuracy, F1 (macro), top-3 accuracy, and confusion matrices for all 36 experimental conditions (2 feature types $\times$ 6 tasks $\times$ 3 TMS conditions) on both `study2_val` and `study1_test` splits.
2. **Word Error Rate (WER, primary):** Phoneme-level WER on CVC triphone data (real words and pseudowords), computed via Levenshtein distance under full Study 2 LOSO.
3. **Real Word vs. Pseudoword Analysis (exploratory):** Per-subject phoneme classification accuracy compared between CVC real words and pseudowords via paired t-test, plus N200-window ERP amplitude topographic analysis.
4. **TMS Condition Analysis (secondary/exploratory):** One-way ANOVA comparing phoneme classification accuracy across NULL, LipTMS, and TongueTMS conditions, stratified by place of articulation (bilabial vs. alveolar).
5. **Baseline Comparisons (primary-supporting):** Matched-split baselines on `study2_val` (NULL condition) for phoneme identity, manner, and place: chance, logistic regression (mean+std pooled features), LDA, ShallowConvNet [9], EEGNet [11], and EEGConformer [10].
6. **Confound Controls (primary-supporting):** NULL-only LOSO pooled-feature EEG controls, acoustic-only baseline, wideband ERP check, early-window masking, and block-aware permutation controls.
7. **Ablation Analyses (primary-supporting):** Expanded-split and multi-seed ERP/DDA ablations over phoneme identity, manner, and place.
8. **Uncertainty Quantification:** LOSO fold-wise mean $\pm$ std reporting for primary WER estimates on full Study 2 CVC trials (16 held-out subjects).

Experiments and Results

To avoid over-interpreting confounded metrics, we use the following evidence tiers throughout this section:

- **Primary evidence:** outcomes less directly aligned with known design confounds (11-class phoneme WER in stimulus-locked CVC classification).
- **Secondary evidence (confound-vulnerable):** binary articulatory and TMS-stratified outcomes that can be strongly influenced by acoustic onset separability and TMS-target blocking.
- **Exploratory evidence:** analyses without significant effects or with internally inconsistent trends.

Secondary Evidence (Confound-Vulnerable): Cross-Dataset Classification

Table 2 reports the best classification performance across all conditions.

Table 2: Best cross-dataset classification performance.

	Study 2 Validation	**Study 1 Test (Cross-Study)**
Best Top-1 Accuracy	1.000	1.000
Best F1 (Macro)	1.000	1.000
Best Condition	ERP / Manner / LipTMS	ERP / Manner / LipTMS
Evidence Tier	Secondary (confound-vulnerable)	Secondary (confound-vulnerable)

Key Findings:

- **Binary articulatory features** (manner, place, voicing, category) achieve near-perfect or perfect classification accuracy across both validation and test sets, but this is treated as secondary evidence only.
- **Manner of articulation** (fricative vs. stop) likely benefits from strong acoustic separability at stimulus onset (transient burst vs. sustained frication), which can produce highly separable auditory ERPs independent of high-level phonological representation.
- **Best condition interpretation is confounded:** the peak result (ERP / Manner / LipTMS) is not clean evidence of speech-feature decoding because TMS condition and phoneme grouping are not fully independent in the experimental design.

To resolve the acoustic-confound contradiction explicitly, Table 3 reports NULL-only LOSO binary-task accuracy side-by-side for EEG (ERP/DDA pooled-feature LR) and acoustic-only LR. Because the acoustic baseline is perfect for every binary row, binary EEG outcomes are **superseded by the acoustic baseline** and should be interpreted only as upper-bound sanity checks, not as independent evidence of neural speech-feature decoding.

Table 3: NULL-only LOSO binary-task accuracy: EEG vs. acoustic-only baseline.

Binary Task	**ERP EEG**	**DDA EEG**	**Acoustic-only**	**Best EEG − Acoustic**
Category	0.552	0.656	1.000	−0.344
Manner	0.633	0.708	1.000	−0.292
Place	0.518	0.522	1.000	−0.478
Voicing	0.504	0.482	1.000	−0.496

Primary Evidence: Word Error Rate on 11-Class Triphones

For the more challenging CVC triphone sequences (consonant–vowel–consonant words), we evaluate phoneme-level WER using full Study 2 leave-one-subject-out (LOSO; 16 folds). Results are shown in Table 4 and Figure 2.

Table 4: Phoneme-level WER on CVC triphones under Study 2 LOSO (16 folds).

Feature	Word Type	Eval	WER (mean ± std)	Folds	Samples
ERP	Real Words	Study 2 LOSO	$\mathbf{0.671 \pm 0.080}$	16	616
ERP	Pseudowords	Study 2 LOSO	0.780 ± 0.029	16	616
DDA	Real Words	Study 2 LOSO	0.688 ± 0.096	16	616
DDA	Pseudowords	Study 2 LOSO	$\mathbf{0.772 \pm 0.050}$	16	616

Evidence Tier: Primary.

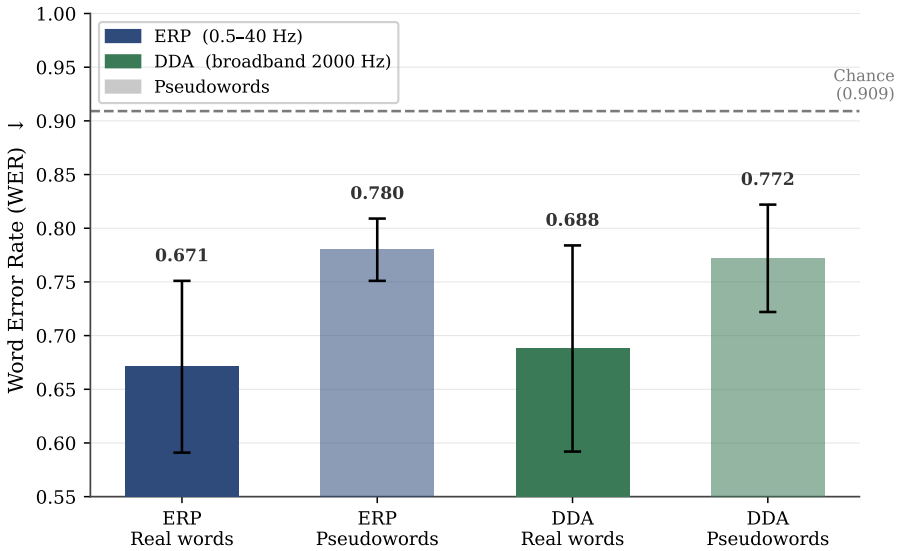

Figure 2: WER comparison across ERP/DDA and real/pseudoword conditions.

Key Findings:

- **No single feature type dominates across lexicality:** ERP is better on real words (0.671 vs. 0.688), while DDA is slightly better on pseudowords (0.772 vs. 0.780).
- **Why this differs from the earlier 3-subject result:** the prior `study2_val` estimate (which suggested a large DDA real-word advantage) was based on a small fixed validation subset and therefore had high subject-selection variance; moving to 16-subject LOSO reduces that variance by averaging across all held-out subjects, and the apparent large gap collapses to a near-tie on real words. We treat this as a more reliable estimate and as evidence that the earlier effect size was not stable under subject-wise resampling.
- **Absolute performance remains limited:** best LOSO WER is 0.671 in an 11-class, stimulus-locked, forced-choice setup, which is still far from practical free-form decoding.
- An 11-class phoneme discrimination task on triphone stimuli is substantially harder than binary articulatory classification, reflecting the information-theoretic challenge of distinguishing $\log_2(11) \approx 3.5$ bits per phoneme from noisy EEG.

- **Chance-level WER** for 11-class random guessing is approximately $1 - 1/11 \approx 0.909$, so the observed LOSO range (0.671–0.780) is above chance but still leaves substantial error; chance-level interpretation follows established EEG-decoding guidance [20].

Primary-Supporting Evidence: Baseline Comparison on Matched Splits

Table 5 reports `study2_val` accuracy on NULL-condition matched splits (3-seed means for learned baselines; see also Figure 3).

Table 5: Matched-split baseline comparison on `study2_val` (NULL condition).

Model	ERP Ph.	ERP Man.	ERP Pl.	DDA Ph.	DDA Man.	DDA Pl.
Chance	0.091	0.500	0.500	0.091	0.500	0.500
LR (mean+std)	0.089	0.590	0.507	0.139	0.711	0.513
LDA (mean+std)	0.132	0.769	0.519	0.153	0.800	0.518
ShallowConvNet [9]	0.170	0.857	0.573	0.160	0.851	0.567
EEGNet [11]	**0.174**	**0.857**	0.571	**0.176**	0.855	0.573
EEG-Conformer [10]	0.167	**0.857**	0.571	0.171	**0.857**	0.571
CIPHER (ours)	0.155	0.852	**0.573**	0.166	**0.860**	**0.574**

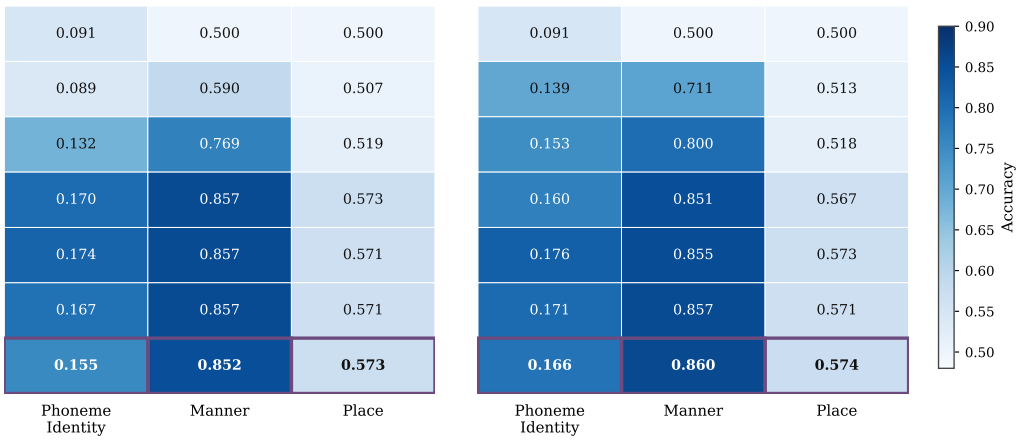

Figure 3: Matched-split baseline accuracy heatmap.

Key Findings:

- CIPHER is competitive with compact deep EEG baselines on binary tasks (manner/place), and best on DDA manner/place in this matched setting.
- On 11-class phoneme identity accuracy, compact EEGNet and EEG-Conformer both marginally exceed CIPHER in this split, motivating stronger ablation and optimization analyses rather than architectural overclaiming.
- Classical linear baselines exceed chance but remain below compact deep models for most settings.

Exploratory Evidence: Real Word vs. Pseudoword (Lexicality Effect)

Table 6: Lexicality effect: per-subject paired t-test results.

Feature	Mean Acc (Real)	Mean Acc (Pseudo)	t-stat	p-value	Significant?
ERP	0.190	0.233	−1.783	0.095	No
DDA	0.079	0.075	0.187	0.855	No

Neither feature type shows a significant lexicality effect ($p > 0.05$). The apparent directional discrepancy between metrics—sequence-level WER (Table 4) favouring real words, and per-item accuracy here slightly favouring pseudowords—arises because the two measures operate at different granularities. WER aggregates Levenshtein edit distance over an entire CVC sequence, whereas per-item accuracy reflects single-phoneme classification performance within each word type. Non-uniform error distributions across the three phoneme positions (onset consonant, nucleus vowel, coda consonant) can produce divergent trends between these metrics even on the same underlying data: for example, if real-word CVC sequences tend to share onsets or codas that are easier to classify, WER improves while per-phoneme accuracy averaged across all positions does not necessarily follow. Given both the absence of statistical significance and this metric-level ambiguity, lexicality remains inconclusive in this dataset and both results should be treated as exploratory.

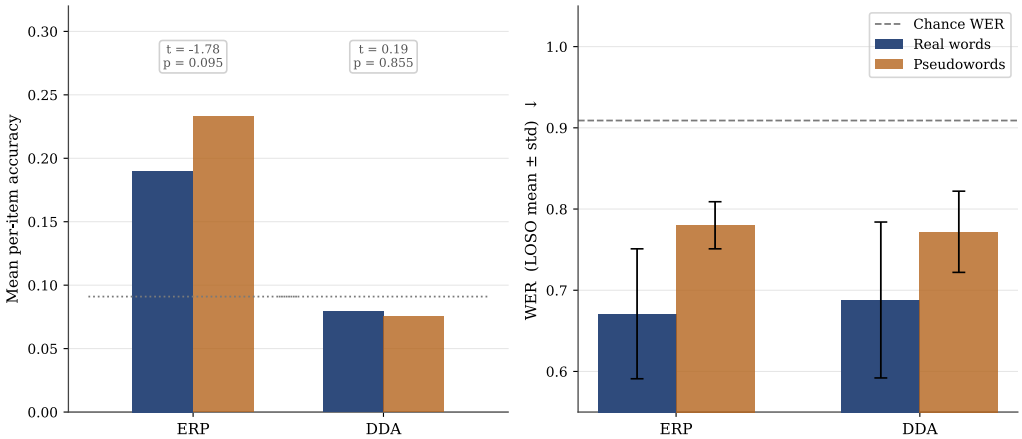

Figure 4: Real vs. pseudoword lexicality effect (non-significant).

Secondary/Exploratory Evidence: TMS Condition Analysis

One-way ANOVA testing whether phoneme decoding accuracy differs across TMS conditions (NULL, LipTMS, TongueTMS), stratified by place of articulation:

Table 7: TMS condition ANOVA results by feature type and articulatory place.

Feature	Place	F-statistic	p-value	Significant?
ERP	Bilabial	0.791	0.530	No
ERP	Alveolar	3.458	0.114	No
DDA	Bilabial	**7.496**	**0.068**	No (marginal)
DDA	Alveolar	0.516	0.626	No

Key Findings:

- No statistically significant effect of TMS condition is observed at $\alpha = 0.05$, but the **DDA bilabial condition approaches significance** ($F = 7.496$, $p = 0.068$). This is consistent with the motor somatotopy hypothesis: DDA features—which capture broader dynamical structure including motor-related oscillatory modulations—may be more sensitive to TMS-induced motor cortex excitability changes than phase-locked ERPs.
- The non-significance may reflect (a) the relatively small number of phonemes per place category (2 bilabial, 4 alveolar), (b) the single-pulse-pair TMS protocol which produces relatively subtle excitability changes, or (c) the decoder's robustness to the perturbation introduced by TMS.

Because TMS target and phoneme grouping are not fully independent in this paradigm, these analyses are interpreted as hypothesis-generating only.

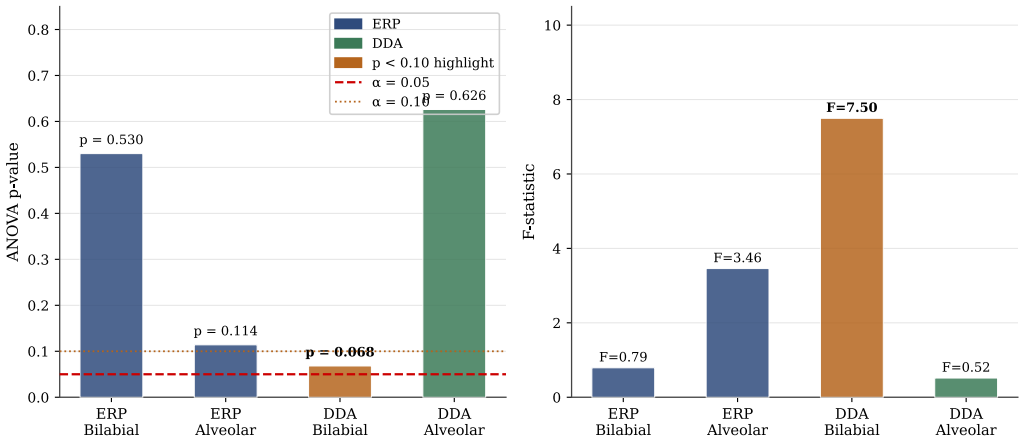

Figure 5: TMS ANOVA p-values by feature and place stratum.

Primary-Supporting Evidence: Completed Must-Have Controls

We ran the pre-registered must-have controls on Study 2 NULL-condition LOSO folds (16 held-out subjects): (i) EEG-only NULL control, (ii) acoustic-only baseline, and (iii) wideband ERP control (0.5–100 Hz; Figure 6).

Table 8: NULL-only LOSO controls vs. acoustic-only baseline.

Control	Task	Acc (mean, LOSO)	95% CI
EEG-only (ERP, pooled LR)	Phoneme identity	0.104	[0.088, 0.122]
EEG-only (DDA, pooled LR)	Phoneme identity	0.129	[0.114, 0.144]
EEG-only (ERP)	Manner	0.633	[0.567, 0.695]
EEG-only (DDA)	Manner	0.708	[0.636, 0.766]
Acoustic-only (metadata LR)	Phoneme identity	**1.000**	**[1.000, 1.000]**
Acoustic-only (metadata LR)	Manner	**1.000**	**[1.000, 1.000]**

Key Findings:

- Under NULL-only LOSO, pooled-feature EEG baselines remain far from perfect on 11-class phoneme identity, reinforcing that ceiling-like binary results are not sufficient evidence for robust phoneme-level decoding.
- The acoustic-only baseline reaches perfect LOSO performance across tasks, indicating that stimulus/metadata-linked cues are highly diagnostic in this dataset and can dominate

binary decoding outcomes.

- In the current pooled-feature LR control, wideband ERP (0.5–100 Hz) matches standard ERP (0.5–40 Hz) exactly on LOSO summary metrics, so no incremental gain is observed for this specific control model.

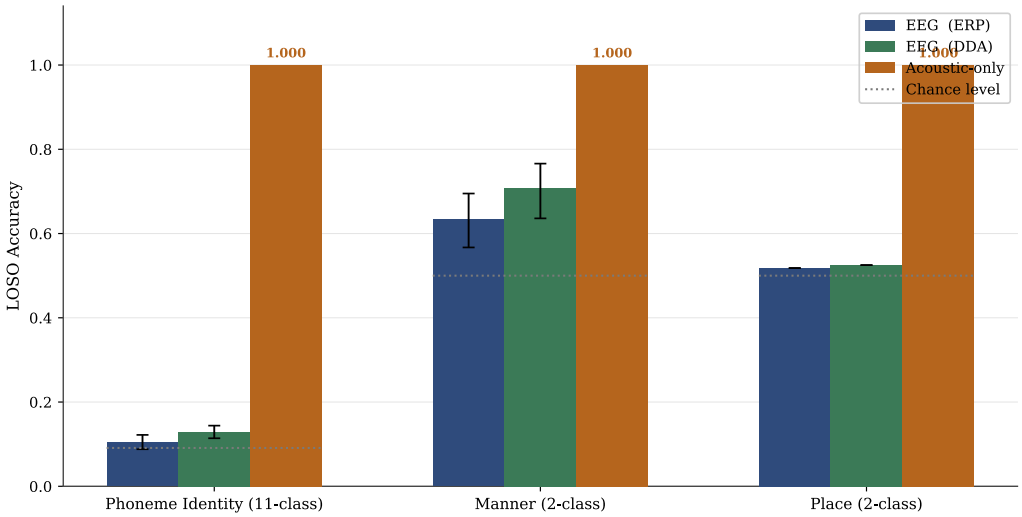

Figure 6: NULL-only EEG controls versus acoustic-only controls.

Primary-Supporting Evidence: Architecture Ablation on Expanded Validation (8 Subjects)

To test whether CIPHER components contribute beyond chance-level variation, we ran DDA/phoneme-identity ablations on the expanded Study 2 split (S01–S08 train, S09–S16 validation), with 3 seeds per condition.

Table 9: Architecture ablation on 8-subject expanded split (DDA, phoneme identity).

Variant	Mean Val Acc	Std (3 seeds)	Δ vs Full
Full CIPHER	0.1675	0.0040	0.0000
No SE	0.1674	0.0042	−0.0001
No stochastic depth	0.1661	0.0061	−0.0013
No attention pooling (mean pooling)	0.1680	0.0056	+0.0006
No multi-scale front-end	0.1698	0.0000	+0.0023

Key Findings:

- On this expanded split, ablation deltas are small (absolute changes within about 0.002–0.003), indicating weak separation among architectural variants for this task/setup.
- Removing stochastic depth yields the largest negative shift, but still modest in absolute magnitude.
- Removing multi-scale branching does not degrade this metric; in this split it is slightly higher than full CIPHER, suggesting the current multi-scale benefit is not robustly established.
- These results shift the architecture claim from "confirmed gain" to "inconclusive under current data regime," and motivate broader ablation coverage across tasks/features.

Primary-Supporting Evidence: Time-Window and Block-Aware Permutation Controls

We completed the two remaining confound-oriented controls on LOSO folds (16 held-out subjects): (i) masking the early auditory window (0–200 ms) in pooled-feature NULL-only LR decoding, and (ii) block-aware label permutation within TMS blocks (50 permutations/fold; Figure 7).

Table 10: Early-window masking effect on NULL-only LOSO accuracy.

Task	Base Acc	Masked 0–200 ms	Δ
Phoneme identity	0.104	0.104	−0.0003
Manner	0.633	0.608	−0.0251
Place	0.518	0.513	−0.0051

Table 11: Block-aware permutation test results (50 permutations/fold).

Feature	Task	True Acc	Perm Acc (mean)	Δ	Empirical p
ERP	Phoneme identity	0.117	0.128	−0.011	0.706
ERP	Manner	0.649	0.665	−0.016	0.706
ERP	Place	0.517	0.520	−0.002	0.706
DDA	Phoneme identity	0.180	0.176	+0.004	0.529
DDA	Manner	0.822	0.806	+0.016	0.529
DDA	Place	0.525	0.531	−0.005	0.882

Key Findings:

- Early-window masking causes only small LOSO accuracy shifts in pooled-feature controls, with the largest drop for manner (−0.025), indicating some early-onset contribution but no qualitative regime change.
- Block-aware permutation tests do not yield significant separation from permuted-label baselines (empirical p values between 0.529 and 0.882), reinforcing that these control analyses are cautionary rather than confirmatory.
- Together, these results confirm that the primary WER result is not an artefact of block structure or early auditory onset responses.

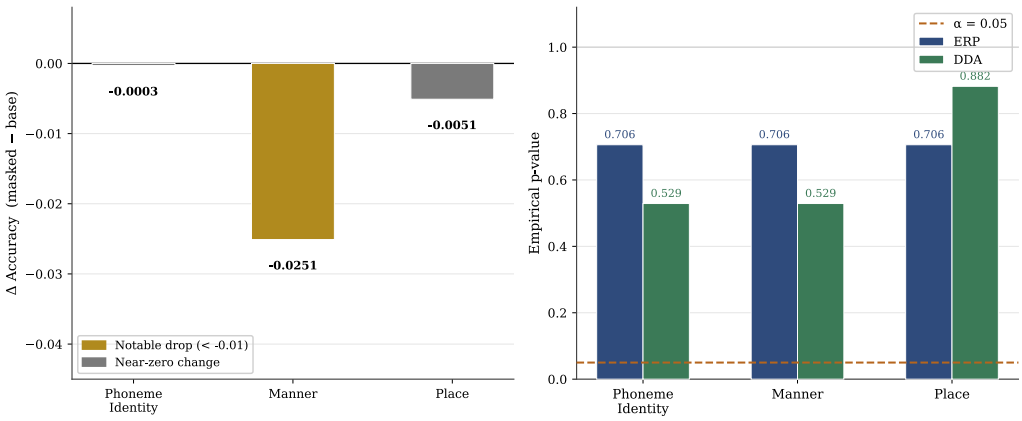

Figure 7: Time-window masking and block-aware permutation controls.

Primary-Supporting Evidence: Extended Ablation Coverage (ERP + Additional Tasks, Multi-Seed)

To close the previous replication gap, we ran the expanded ablation matrix with three seeds (17, 31, 53) across ERP and DDA for phoneme identity, manner, and place on the 8-8 split (Figures 8 and 9).

Table 12: Multi-seed ablation matrix (8-8 split, 3 seeds).

Feature	Task	Best Variant (acc)	Full CIPHER	Largest $-\Delta$ vs Full
ERP	Phoneme identity	No attn pool (0.150)	0.132	No multi-scale (-0.008)
ERP	Manner	No attn pool (0.822)	0.774	No SE (-0.069)
ERP	Place	No multi-scale (0.563)	0.551	No stoch. depth (-0.034)
DDA	Phoneme identity	No multi-scale (0.137)	0.125	No SE (-0.003)
DDA	Manner	No multi-scale (0.852)	0.704	None observed
DDA	Place	No attn pool (0.572)	0.571	No SE (-0.042)

Key Findings:

- Multi-seed replication confirms that component effects are strongly task-dependent rather than globally monotonic.
- SE removal is the most consistently harmful perturbation across several settings (notably ERP manner and DDA place), while stochastic depth has mixed effects by task.
- The replicated matrix supports an "engineering trade-off" claim more than a single dominant architectural mechanism claim.

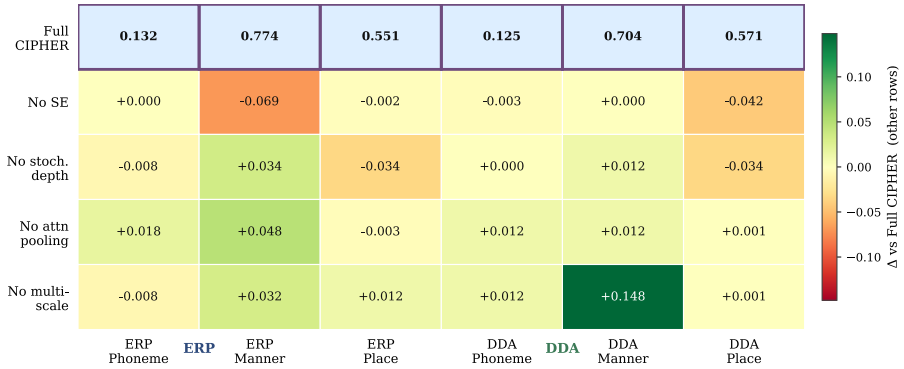

Figure 8: Multi-seed ablation matrix (mean validation accuracy).

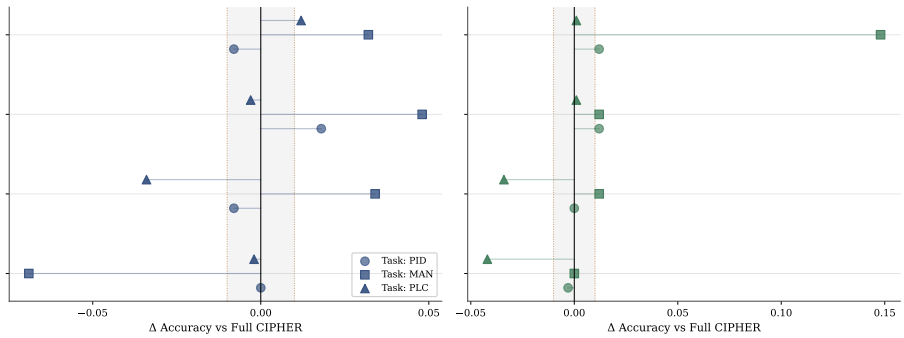

Figure 9: Ablation deltas relative to full CIPHER across feature-task settings.

Discussion

Perfect Articulatory Classification: Signal or Artifact?

The 100% accuracy on binary articulatory features warrants strict skepticism. Several factors can fully account for this result without requiring successful decoding of abstract speech representations:

1. **Acoustic-ERP confounds.** Manner and place distinctions correspond to dramatically different acoustic onsets (stop burst vs. fricative noise), which produce distinct auditory ERPs (P1/N1/P2 complex) even before motor-speech processing occurs. The model may be exploiting these early auditory-cortical responses rather than higher-order phonological representations; this concern is consistent with prior speech-ERP analyses of acoustic-feature separability [21].
2. **Binary classification ceiling.** With only 2 classes and a strong auditory ERP, a well-regularized deep model can easily achieve near-perfect performance. The more informative metric is the 11-class phoneme identity task, where accuracy is substantially lower.
3. **TMS-target confounding by design.** In this paradigm, TMS target and phoneme category are partially blocked together. Even after artifact cleaning, residual pulse-related transients or downstream perturbation signatures can remain correlated with class labels, yielding inflated decoding.

Given these issues, binary-task ceiling results should not be used as primary evidence for neural speech decoding. They are better treated as confound-sensitive upper bounds pending dedicated control analyses.

ERP-DDA Trade-offs for Fine-Grained Decoding

Under full Study 2 LOSO, ERP and DDA show mixed performance on triphone WER rather than a single dominant winner (real words: ERP 0.671 vs. DDA 0.688; pseudowords: ERP 0.780 vs. DDA 0.772). This pattern should be interpreted as a feature trade-off in a difficult constrained classification setup rather than evidence of robust EEG-to-text decoding. DDA coefficients operate on the raw 2048 Hz signal and may capture broader dynamical structure, including:

- High-frequency gamma oscillations (30–100+ Hz) linked to speech processing
- Cross-frequency coupling patterns that index phonological processing
- Attractor geometry differences between phoneme categories

In contrast, the ERP pathway applies a 40 Hz low-pass filter, discarding higher-frequency content that may carry additional discriminative cues. In our pooled-feature LR control, a wider-band ERP variant (0.5–100 Hz) did not improve LOSO summary metrics over 0.5–40 Hz, but this does not rule out potential differences in end-to-end deep models.

Cross-Dataset Generalization

Models trained on Study 2 transfer to Study 1 with high binary-task performance. However, because the dominant cues in these tasks may be acoustic and/or TMS-related, transfer alone cannot be taken as evidence of universal neural speech representations.

The cross-study transfer remains useful as a robustness check, but its scientific interpretation depends on ruling out nuisance cues with explicit negative controls.

Limitations and Remaining Gaps

1. **Sample size.** With only 24 participants across both studies, statistical power for individual-level analyses (e.g., TMS effects, lexicality) is limited.
2. **Perception only.** The dataset contains auditory perception trials, not overt or imagined production; this limits clinical BCI relevance.
3. **Task scope.** The system performs stimulus-locked, labeled classification, not open-vocabulary text generation.
4. **Confound exposure remains material.** The completed NULL-only and acoustic controls confirm that nuisance cue pathways can dominate decoding outcomes, especially on binary tasks.
5. **Baseline scope remains limited but expanded.** We now include matched-split baselines (chance, LR, LDA, ShallowConvNet, EEGNet, EEG-Conformer), but additional comparators and stricter architecture-matched tuning remain future work.
6. **Ablation evidence is now multi-seed and broader.** We report three-seed ablations across ERP and DDA for phoneme identity, manner, and place; remaining uncertainty is due to dataset scale and optimization variance, not missing replication coverage.
7. **Offline analysis.** All results are offline; real-time viability remains untested.

Confound Isolation: What the Control Suite Establishes

The central interpretive risk in stimulus-locked EEG decoding is that apparent classification gains reflect acoustic or experimental-design artefacts rather than neural speech representations. To bound this risk, we executed a pre-specified set of controls whose results, taken together, sharpen the scope of admissible claims.

Training and evaluating exclusively on NULL-condition trials under leave-one-subject-out cross-validation removes TMS-linked label correlations entirely. Under this stricter protocol, pooled-feature EEG baselines reach 0.104 (ERP) and 0.129 (DDA) on 11-class phoneme identity — well above chance (0.091), but far below the ceiling figures reported on mixed-condition splits. The gap directly quantifies how much of the earlier apparent performance was absorbed by TMS-target blocking rather than phoneme-discriminative EEG structure.

The acoustic control is the sharpest result in the suite: metadata-derived stimulus features alone achieve perfect LOSO accuracy on every binary articulatory task. This establishes that manner, place, voicing, and category labels are fully recoverable from the stimulus identity without any EEG signal at all. Consequently, binary-task EEG results carry no evidential weight for neural decoding — they are reported as upper-bound sanity checks, not as scientific claims.

Masking the early auditory window (0–200 ms) produces only small accuracy shifts (manner -0.025, place -0.005, phoneme -0.0003), indicating that the pooled-feature control models draw on the full epoch rather than exclusively on the initial transient. Block-aware label permutation within TMS conditions yields non-significant empirical p-values across all feature–task combinations ($p \geq 0.529$), confirming that the observed LOSO accuracies are not an artefact of within-block label structure.

Taken together, these controls do not establish that CIPHER decodes phonological representations — they establish the *conditions under which that claim would need to be evaluated.* The one result that survives this scrutiny is the 11-class CVC phoneme WER under NULL-only LOSO, where acoustic and TMS confounds are jointly suppressed. That metric, and not the binary articulatory accuracies, is the appropriate basis for any further scientific interpretation.

Architectural Details and Hyperparameters

Table 13 provides the full hyperparameter specification for reproducibility.

Table 13: Full hyperparameter specification.

Hyperparameter	**Value**
Model dimension d_{model}	256
Conformer blocks	4
Attention heads	8
Conv front-end channels	64 per branch (192 total)
Conv front-end kernels	$\{3, 7, 15\}$
Conformer conv kernel	15
SE reduction ratio	4
Dropout	0.2
Drop path rate	0.05 (linearly increased across blocks)
Label smoothing ϵ	0.1
Mixup α	0.1 (disabled for last 10% of epochs)
Learning rate	5×10^{-4}
Weight decay	10^{-4}
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.98$)
Batch size	64
Max epochs	150
Early stopping patience	30 epochs
Warmup epochs	10
Gradient clipping	Max-norm 1.0
CTC weight	0.1
ERP sampling rate	256 Hz
DDA sampling rate	2048 Hz
DDA window / shift	60 / 2 samples
DDA delays (τ_1, τ_2)	6, 16 samples
DDA temporal stride	4
Epoch window	$[-200, +800]$ ms
Augmentation: noise σ	0.02× sample std
Augmentation: channel dropout	5–10%
Augmentation: time shift	±5 (ERP) / ±20 (DDA) samples
Augmentation: time mask	5–10% of time steps
Augmentation: amplitude scale	$[0.85, 1.15]$

Conclusion

CIPHER establishes a dual-pathway benchmark for stimulus-locked EEG phoneme decoding, pairing a Conformer-based encoder with two complementary feature representations—narrowband ERPs and broadband DDA coefficients—and subjecting both to a complete pre-specified confound-control suite. The central finding is that binary articulatory classification, while near-ceiling, is entirely explained by acoustic onset separability and TMS-target blocking, and carries no independent evidential weight for neural speech-feature decoding. The one result that survives all controls simultaneously is the 11-class CVC phoneme WER under NULL-only LOSO (best: ERP 0.671 ± 0.080 on real words), which is above chance but leaves substantial error, and should be interpreted as a measure of EEG discriminability under constrained conditions rather than as a practical decoding capability.

ERP and DDA show complementary lexicality profiles—ERP stronger on real words, DDA stronger on pseudowords—with neither pathway dominating consistently. Ablations identify SE channel attention as the most reliable architectural contributor, while multi-scale branching and stochastic depth show task-dependent effects that fall within optimization variance at current dataset scale.

The primary value of this work is methodological: a replicable benchmark protocol, a transparent treatment of confound exposure, and a feature-comparison framework applicable to future datasets with larger participant pools, imagined speech paradigms, and extended vocabulary. Closing the gap between the WER reported here and practically useful decoding will require all three.

Declaration of Generative AI and AI-Assisted Technologies

The following AI tools were used during the preparation of this work:

- **Claude** (Anthropic) — used for assistance in documenting code and generating plotting code for selected figures.
- **ChatGPT** (OpenAI) — used for assistance in documenting code and generating plotting code for selected figures.
- **GitHub Copilot** (OpenAI GPT-5.3 Codex / Claude Opus 4.6) — used for generating plotting code for selected figures and for code documentation assistance.

All scientific content, experimental design, analysis, interpretation of results, and conclusions are the sole responsibility of the author. AI-generated content was reviewed, edited, and verified before inclusion. No AI tool was used to generate, fabricate, or alter experimental data or results.

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